

100 Days of Linux - Day 050

Title: Practical Assignment - Linux Process Management

Today's Goal

Apply everything learned about Linux process management in one realistic troubleshooting workflow. No new commands are introduced today.

Why This Matters

Monitoring, controlling, and troubleshooting processes are everyday tasks for Linux administrators, developers, DevOps engineers, and data engineers.

Commands Covered

ps, top, htop, kill, jobs, fg, bg

Scenario

A data processing job is running on your Linux system. You need to inspect it, monitor resources, manage background jobs, and safely stop processes when required.

Practical Assignment (5 Tasks)

1. Start two long-running commands using 'sleep 300' in separate terminals. Use ps -ef to identify both PIDs.
2. Open top and identify which process is using the most CPU. Exit top cleanly.
3. Open htop (install it if needed). Search for one sleep process, switch to Tree View, then exit htop.
4. Pause one sleep process with Ctrl+Z. Use jobs to verify it, resume it with bg, bring it back with fg, then stop it using Ctrl+C.
5. Start another sleep 300 process. Find its PID with ps and stop it gracefully using kill. If it does not stop, use kill -9. Display your current working directory.

Hints

Complete one task at a time and verify the result after every command using ps or jobs.

Mini Challenge

Run three sleep processes. Monitor them with top and htop, suspend one, background another, terminate one with kill, and verify the final state with ps.

Common Mistakes

Confusing job numbers (%1, %2) with PIDs. Jobs are managed by the shell; kill uses PIDs.

Did You Know?

Most production monitoring tools build on the same process information exposed by the Linux kernel that commands like `ps` and `top` display.

Pro Tip

Always try a graceful shutdown before using forceful termination with `kill -9`.

Answer Key (Instructor Only)

Q1: `sleep 300 && ps -ef | grep sleep`

Q2: `top` then press `q`

Q3: `htop`, `F3` search, `F5` tree, `F10` quit

Q4: `Ctrl+Z`, `jobs`, `bg %1`, `fg %1`, `Ctrl+C`

Q5: `ps -ef | grep sleep && kill PID (or kill -9 PID if required) && pwd`

Homework

Repeat today's workflow until you can confidently inspect, monitor, suspend, resume, and terminate processes without referring to notes.

Next Day Preview

Tomorrow you'll begin the Networking module by learning the `ping` command to test network connectivity.